C. Lose it!

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1300 | Tags: dp, greedy, implementation | https://codeforces.com/problemset/problem/1176/C

You are given an array *a* consisting of *n* integers. Each a_i is one of the six following numbers: 4, 8, 15, 16, 23, 42.

Your task is to remove the minimum number of elements to make this array **good**.

An array of length *k* is called **good** if *k* is divisible by 6 and it is possible to split it into *k*/6 **subsequences** 4, 8, 15, 16, 23, 42.

Examples of good arrays:

- [4, 8, 15, 16, 23, 42] (the whole array is a required sequence);
- [4, 8, 4, 15, 16, 8, 23, 15, 16, 42, 23, 42] (the first sequence is formed from first, second, fourth, fifth, seventh and tenth elements and the second one is formed from remaining elements);
- [] (**the empty array is good**).

Examples of bad arrays:

- [4, 8, 15, 16, 42, 23] (the order of elements should be exactly 4, 8, 15, 16, 23, 42);
- [4, 8, 15, 16, 23, 42, 4] (the length of the array is not divisible by 6);
- [4, 8, 15, 16, 23, 42, 4, 8, 15, 16, 23, 23] (the first sequence can be formed from first six elements but the remaining array cannot form the required sequence).

Input

The first line of the input contains one integer *n* ($1 \le n \le 5 \cdot 10^5$) — the number of elements in *a*.

The second line of the input contains *n* integers $a_1, a_2, \ldots, a_n$ (each a_i is one of the following numbers: 4, 8, 15, 16, 23, 42), where a_i is the *i*-th element of *a*.

Output

Print one integer — the minimum number of elements you have to remove to obtain a **good** array.

Examples

Input
5 4 8 15 16 23

Output
5

Input
12 4 8 4 15 16 8 23 15 16 42 23 42

Output

```
0
```

Input

```
15
4 8 4 8 15 16 8 16 23 15 16 4 42 23 42
```

Output

```
3
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